

MLib Yield Curve Overview

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MHI

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29th February 2016

MLib Analytics Framework

- MHI use the MLib C++ analytics library for curve construction

Yield Curves:

- MLib supports Multi-Yield Curves
- OIS Discounting and CSA support is available
- In Excel a full set of USD curves build in approx 0.6 seconds

- ① Curve Types
- ② Curves
 - OIS Curves
 - Standard Swap Curves
 - Basis Curves
 - CSA Curves
 - FX Forward Invariance
- ③ Build Features
 - Curve Output
 - Interpolation
 - Tension
 - Risk Jacobians
- ④ Curve Building Examples
- ⑤ Curve Build Order

Multi-Curves

MLib supports a full variety of curves including OIS, Standard Swap, Tenor Basis, Xccy Basis and CSA curves. These can be constructed using the below functions

Curve Types:

`mirSetUpOISCurve`
`mirSetUpSwapCurve`
`mirSetUpBasisSwapCurve`

OIS Curves
Standard Swap Curves
Tenor Basis Curves

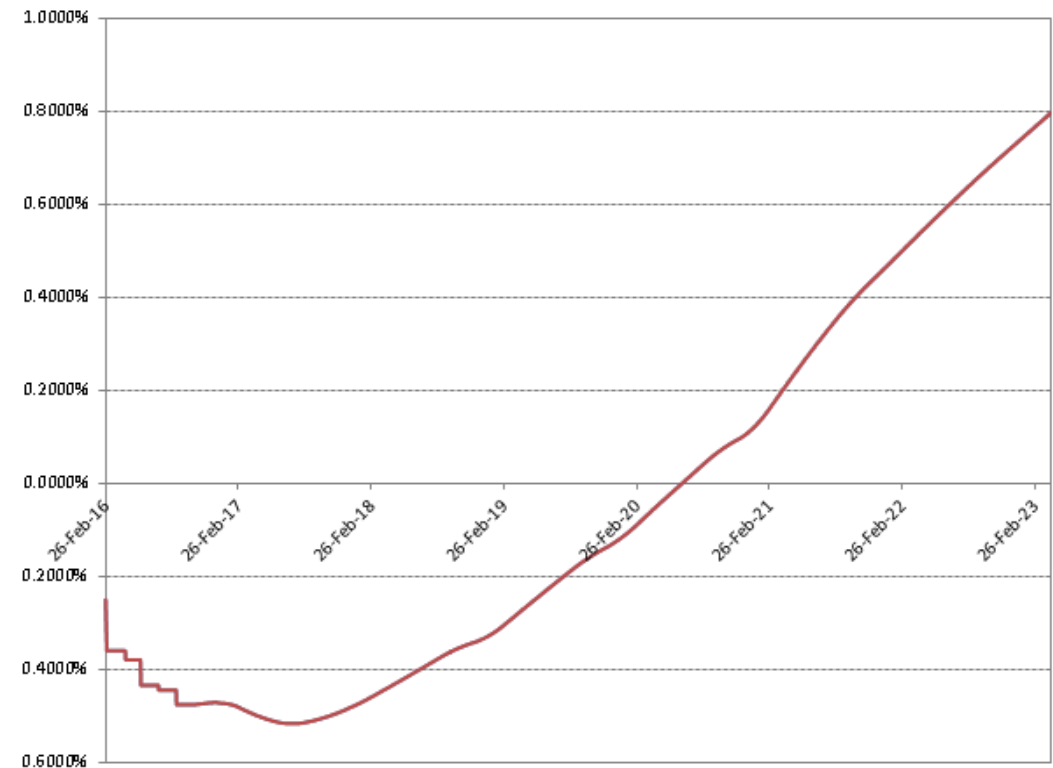
CSA Curves:

`mirSetUpBasisSwapCurve`
`mirSetUpFwdFXConstantCurve`

Xccy Basis Curves
CSA Curves

Calibration Instruments

- OIS Swaps
- Libor-OIS Basis Swaps
- Central Bank Dated Swaps
- e.g. ECB and Fed Fund Swaps

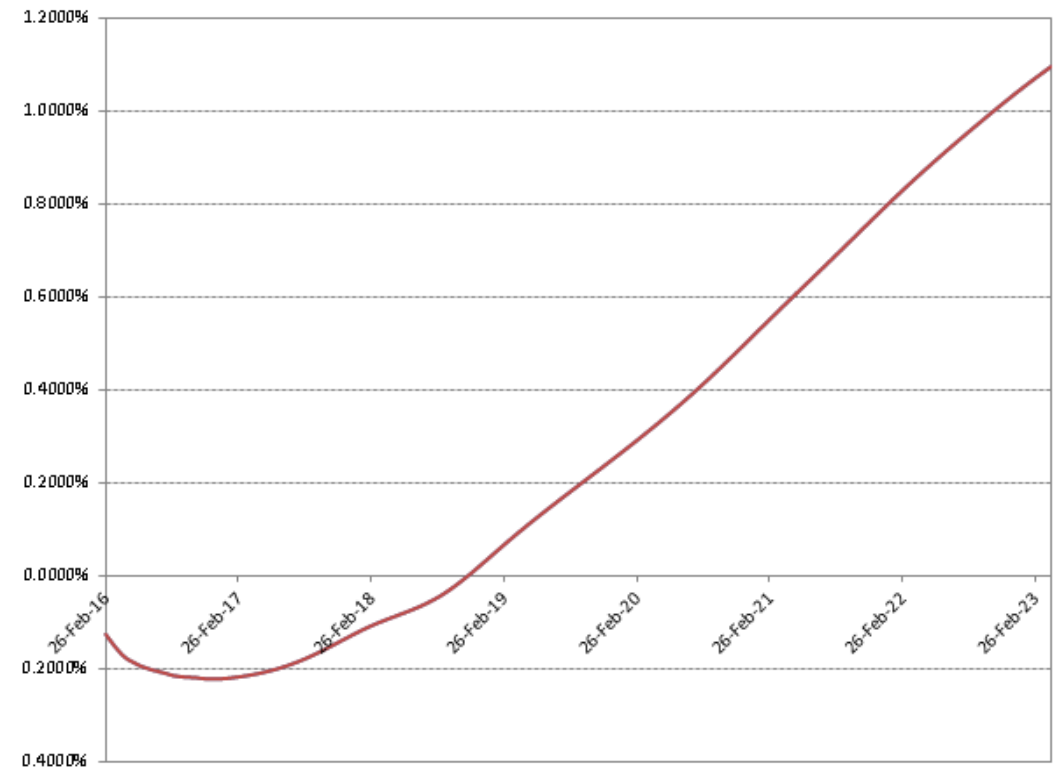


Curve Features

OIS curves are typically constructed using OIS Swaps and Libor-OIS Basis Swaps. The short end is interpolated piecewise constant between Central Bank meeting dates.

Calibration Instruments

- Libor Rates
- FRAs
- Futures
- Swaps

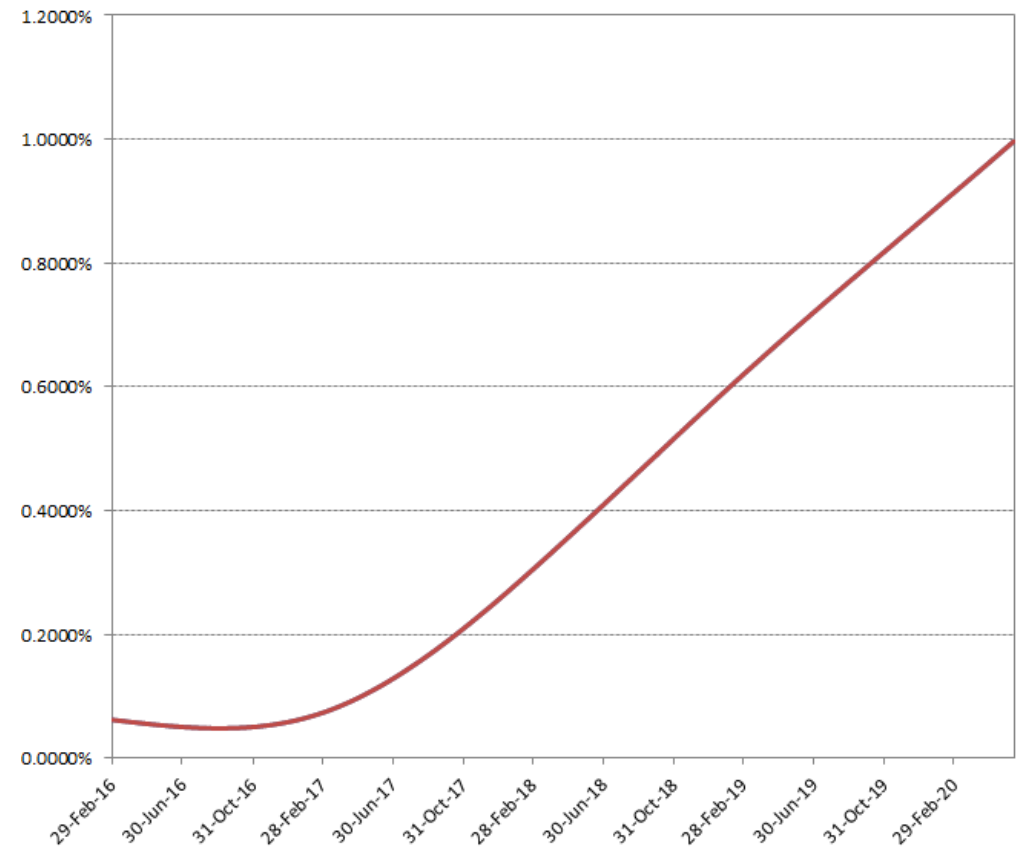


Curve Features

Standard swap curves are using the above instruments. Curves are built using global optimization techniques having an analytical risk Jacobian. Furthermore Tension features are available

Calibration Instruments

- Tenor Basis Swaps
- or Xccy Basis Swaps



Curve Features

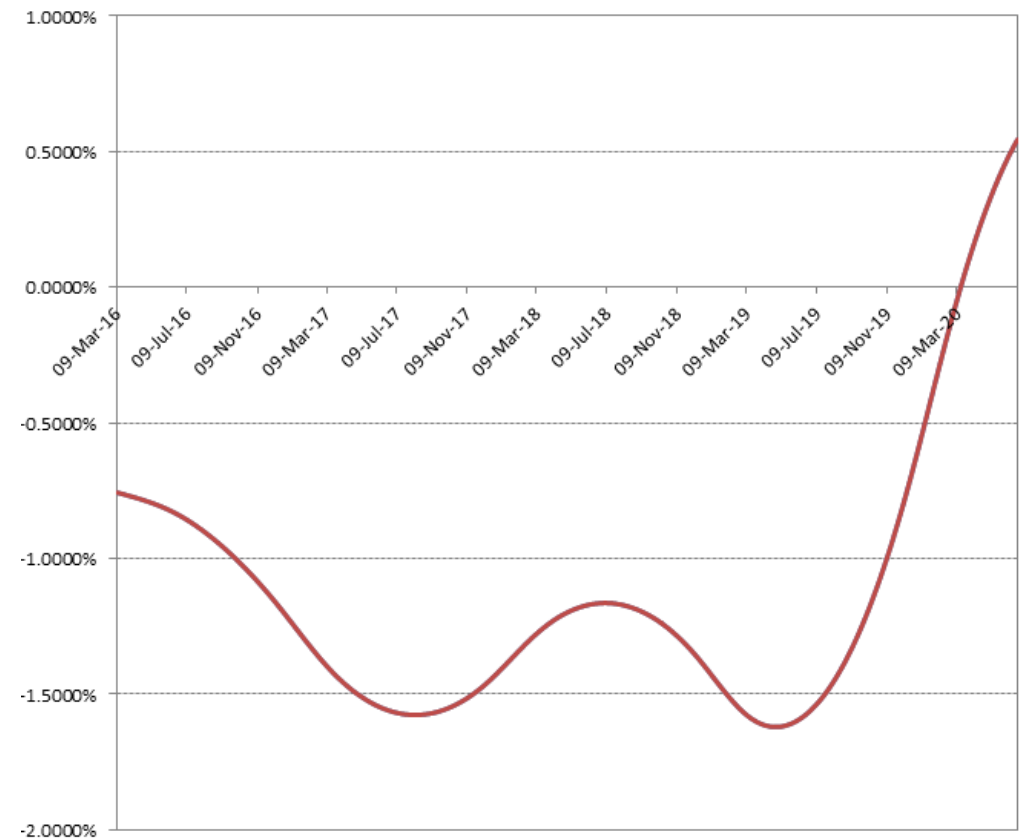
Both Tenor & Xccy Basis Curves are built using bootstrapping / solver techniques

Build Method

- Bootstrapping
- Assumes constant FX Forwards or **FX Forward Invariance**

Calibration Instruments

- Xccy Basis Swaps
- FX Forwards



Curve Features

CSA curves are bootstrapped from Xccy swaps and assume FX Forward invariance, see next slide.

- We assume that FX Forwards are constant
- FX Forwards are fixed assuming a USD CSA

FX Forward Invariance - Example

$$FwdFX(t, T)^{EUR/GBP} = S \frac{P(t, T)^{EUR_USDCSA}}{P(t, T)^{GBP_USDCSA}} = S \frac{P(t, T)^{EUR_GBPCSA}}{P(t, T)^{GBP_GBPCSA}}$$

where t denotes the valuation date, S the FX spot rate, $P(t, T)$ the discount factor at time t for tenor T with $0 \leq t < T$

- From which Discounted CSA Curves can then be implied

Discount CSA Curve - General Formula

$$P(t, T)^{Ccy1_Ccy2CSA} = \left(\frac{P(t, T)^{Ccy1_USDCSA}}{P(t, T)^{Ccy2_USDCSA}} \right) \times \underbrace{P(t, T)^{Ccy2_Ccy2CSA}}_{\text{Ccy2 OIS Discount Factor}}$$

- Curves output consists of a set of Discount Factors & Forwards
- Discount Factors can be retrieved using [mirOutputCurve](#)
- Forwards can be retrieved using [mirGetForwardRate2](#)

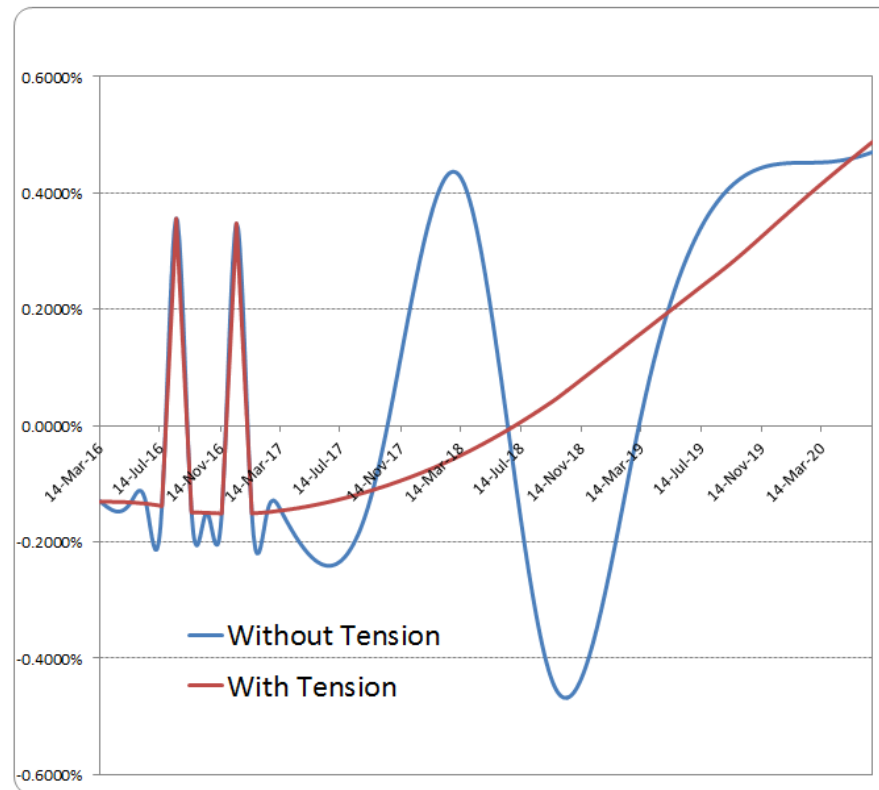
Curve ID	EURYC	Curve ID	EURYC	Curve ID	EURYC	Curve ID	EURYC	Curve ID	EURYC
Curve Name	EUR1ML	Curve Name	EUR3ML	Curve Name	EUR6ML	Curve Name	EUR12ML	Curve Name	EURDF_NOCSA
Description	1ML Projection Curve	Description	3ML Projection Curve	Description	6ML Projection Curve	Description	12ML Projection Curve	Description	Discount Curve No CSA

Term	DiscFact	Term	DiscFact	Term	DiscFact	Term	DiscFact	Term	DiscFact
0.00	1.000000	0.00	1.000000	0.00	1.000000	0.00	1.000000	0.00	1.000000
0.01	1.000045	0.01	1.000030	0.01	1.000010	0.01	1.000002	0.01	1.000020
0.10	1.000390	0.26	1.000732	0.01	1.000014	1.01	1.000227	0.01	1.000031
0.18	1.000748	0.51	1.001452	0.18	1.000229	2.01	1.000561	0.01	1.000041
0.26	1.001073	0.76	1.002198	0.19	1.000235	3.01	0.999824	0.02	1.000051
0.34	1.001412	1.01	1.002990	0.19	1.000241	4.02	0.997055	0.02	1.000061
0.43	1.001777	1.26	1.003859	0.20	1.000247	5.01	0.991868	0.03	1.000092
0.51	1.002123	1.51	1.004757	0.20	1.000253	6.01	0.984171	0.03	1.000102
0.60	1.002473	1.76	1.005634	0.21	1.000258	7.01	0.973544	0.03	1.000113
0.68	1.002840	2.01	1.006449	0.21	1.000264	8.01	0.960571	0.04	1.000123
0.76	1.003201	2.26	1.007196	0.22	1.000269	9.01	0.946044	0.04	1.000133
0.85	1.003606	2.51	1.007842	0.22	1.000274	10.01	0.930846	0.05	1.000164
0.93	1.003970	2.76	1.008378	0.23	1.000280	11.01	0.914870	0.05	1.000174
1.01	1.004328	3.01	1.008800	0.23	1.000285	12.01	0.898666	0.05	1.000184
1.10	1.004785	3.26	1.009115	0.24	1.000290	13.01	0.882585	0.05	1.000195
1.18	1.005145	3.52	1.009302	0.25	1.000295	14.01	0.866796	0.06	1.000205
1.26	1.005549	3.77	1.009338	0.25	1.000300	15.01	0.851590	0.07	1.000236
1.34	1.005958	4.02	1.009220	0.26	1.000305	16.01	0.837089	0.07	1.000246
1.43	1.006382	4.27	1.008935	0.26	1.000310	17.01	0.823492	0.07	1.000256
1.51	1.006808	4.51	1.008479	0.27	1.000315	18.01	0.810540	0.07	1.000266
1.60	1.007246	4.76	1.007848	0.27	1.000320	19.01	0.798253	0.08	1.000277
1.68	1.007641	5.01	1.007037	0.28	1.000325	20.01	0.786558	0.08	1.000307
1.76	1.008047	5.26	1.006041	0.28	1.000330	21.01	0.775425	0.09	1.000318
1.85	1.008473	5.51	1.004835	0.29	1.000335	22.01	0.764878	0.09	1.000328
1.93	1.008865	5.76	1.003435	0.30	1.000339	23.01	0.754799	0.09	1.000338
2.01	1.009224	6.01	1.001833	0.30	1.000344	24.01	0.745093	0.11	1.000389
2.10	1.009636	6.26	0.999968	0.31	1.000349	25.01	0.735804	0.11	1.000400
2.18	1.009977	6.51	0.997886	0.31	1.000354	26.01	0.726826	0.11	1.000410

Current interpolation methods for curve construction:

- Piecewise Constant [fn_stepInterpolation](#)
- Linear [fn_linearInterpolation](#)
- Monotone Convex [fn_monotoneconvexInterpolation](#)
- Spline [fn_splineInterpolation](#)
- Constrained Spline [fn_constrainedsplineInterpolation](#)

- Curve tension can be applied to swap curves
- Can be added to FRAs and / or Futures Instruments
- Used to manage curve over-oscillation & market data shocks



- All curves use optimized using Newton-Raphson like solvers
- First-order risk matrices or **Jacobians** are a by-product

Newton-Raphson Optimization

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

where $f'(x_n)$ is our first order sensitivity

- Curves are being extended to store and use this data
- The Jacobian is useful for analytical risk and speedy curve re-builds

Risk Jacobian

Curves are globally optimized having analytical Jacobian risk metrics, $\mathcal{J}$

$$\underline{L}_{n+1} = \underline{L}_n - \mathcal{J}F(\underline{L}_n)$$

where $\underline{L}$ is our Libor vector, $F(\underline{L})$ the calibration pricing kernel and $\mathcal{J} = F'(\underline{L}_n)^{-1}$ is the Jacobian or sensitivity matrix

mirSetUpOISCurve (CurveCollection, StaticDataTable, CurveBuildProperties, OISConventions, OISMktData, IndexName, OISFixingTable, OISBasisMktData, SwapConventions, SwapMktData, OISBasisConventions)

mirSetUpOISCurve

mirSetUpOISCurve	OIS OIS Curve has been set (0.04700 sec)
Curve Names	
CurveCollection	EURYC
StaticDataTable	OIS
IndexName(s)	OIS:EURDF

Curve Build Properties

Curve Build Properties		Interpolation Options
AsOfDate	19-Feb-16	fn_LinearInterpolation
Currency	EUR	fn_StepInterpolation
yieldgen.interpolation	fn_constrainedsplineInterpolation	fn_MonotoneConvexInterpolation
		fn_SplineInterpolation
		fn_ConstrainedSplineInterpolation

Overnight Indexed Swaps

OIS Conventions		OIS Par Rates					OIS Fixing Table	
calendar	TGT	Term	Rate	StartDate	EndDate	Use	Date	Rate
daycount	ACT/360	ON	-0.244%			TRUE		
frequency	Annual	EUSF1A	-0.369%	10-Mar-16	21-Apr-16	TRUE		
resetlag	2d	EUSF2A	-0.392%	21-Apr-16	02-Jun-16	TRUE		
slidingrule	MOD_FOLLOWING	EUSF3A	-0.445%	02-Jun-16	21-Jul-16	TRUE		
iseomroll	TRUE	EUSF4A	-0.453%	21-Jul-16	08-Sep-16	TRUE		
eomday	31	EUSF5A	-0.476%	08-Sep-16	20-Oct-16	TRUE		
generatemethod	dailycompounding	12M	-0.433%			TRUE		
firstrate	ShortMarket	18M	-0.448000%			TRUE		
shorttermconvention	arithmeticaverage	2Y	-0.448%			TRUE		
longtermconvention	LOBASIS	3Y	-0.409%			TRUE		
longterm	3Y	4Y	-0.337%			TRUE		
longterm.generatemethod	dailyaveraging	5Y	-0.249%			TRUE		
		6Y	-0.140%			TRUE		
		7Y	-0.024%			TRUE		
		8Y	0.093%			TRUE		
		9Y	0.205%			TRUE		
		10Y	0.305%			TRUE		

Libor-OIS Basis Swaps

Libor-OIS Tenor Basis Conventions		Libor-OIS Spreads		Libor Swap Conventions		Libor Swap Rates	
Calendar	TGT	Term	Rate	calendar	TGT	Term	Rate
Daycount	ACT/360	2Y	0.155%	daycountfix	30/360	2Y	-0.295%
Frequency	Quarterly	3Y	0.159%	frequencyfix	Annual	3Y	-0.254%
SlidingRule	MOD_FOLLOWING	4Y	0.162%	slidingrule	MOD_FOLLOWING	4Y	-0.178%
		5Y	0.165%			5Y	-0.085%
		6Y	0.163%			6Y	0.023%
		7Y	0.162%			7Y	0.140%
		8Y	0.161%			8Y	0.258%
		9Y	0.160%			9Y	0.370%
		10Y	0.159%			10Y	0.471%

mirSetUpOISCurve (CurveCollection, StaticDataTable, CurveBuildProperties, MoneyMktConventions, LiborConventions, LiborMktData, SwapConventions, SwapMktData, FRAConventions, FRAMktData3M, FRAMktData6M, FuturesConventions, FuturesMktData, [QuantoConventions], [QuantoMktData], IndexName, [TradeDiscountCurve])

mirSetUpSwapCurve

mirSetUpSwapCurve	STD Swap Curve has been set (0.01600 sec)
Curve Loaded	TRUE
Curve Names	
CurveCollection	EURYC
StaticDataTable	STD
IndexName(s)	STD:EUR6ML

Curve Build Properties

Curve Build Properties		Interpolation Options
AsOfDate	19-Feb-16	fn_LinearInterpolation
Currency	EUR	fn_StepInterpolation
future.interpolation	fn_MonotoneConvexInterpolation	fn_MonotoneConvexInterpolation
interpolation	fn_MonotoneConvexInterpolation	fn_SplineInterpolation
yieldgen.interpolation	fn_MonotoneConvexInterpolation	fn_ConstrainedSplineInterpolation
isfrause	TRUE	
isfutureuse	FALSE	
Frequency	Semi-Annual	
dfcurvename	OIS	

Example Swap Curve III

Futures Instruments

Futures Conventions

calendar	NYB
daycount	Act/360
resetlag	2
slidingrule	MOD_FOLLOWING
MeanReversion	3.00%
ApplyTension	TRUE
TensionGap	2

Futures Data

Term / Ticker	StartDate	EndDate	Rate	Volatility
ED2 COMB Comdty	21-Sep-16	21-Dec-16	99.1700	0.900%
ED3 COMB Comdty	21-Dec-16	15-Mar-17	99.0850	0.900%
ED4 COMB Comdty	15-Mar-17	21-Jun-17	99.0250	0.900%
ED5 COMB Comdty	21-Jun-17	20-Sep-17	98.9600	0.900%
ED6 COMB Comdty	20-Sep-17	20-Dec-17	98.8850	0.900%
ED7 COMB Comdty	20-Dec-17	21-Mar-18	98.8100	0.900%
ED8 COMB Comdty	21-Mar-18	20-Jun-18	98.7500	0.900%
ED9 COMB Comdty	20-Jun-18	19-Sep-18	98.6800	0.900%
ED10 COMB Comdty	19-Sep-18	19-Dec-18	98.6150	0.900%
ED11 COMB Comdty	19-Dec-18	20-Mar-19	98.5400	0.900%
ED12 COMB Comdty	20-Mar-19	19-Jun-19	98.4850	0.900%

Swap Instruments

Swap Conventions

Calendar	LNB:NYB
DayCount	30/360
DayCountFix	30/360
DayCountFloat	ACT/360
Frequency	Semi-Annual
FrequencyFloat	Quarterly
Interpolation	fn_MonotoneConvexInterpolation
IsEomRoll	FALSE
IsNewtonraphson	TRUE
IsTimeInterpolation	TRUE
Resetlag	2
SlidingRule	MOD_FOLLOWING
issimultaneousseq	TRUE
optimizemethod	NL2SOL
Epsilon	1.00E-09
GradientEpsilon	1.00E-15
Delta	1.00E-10
MaxLoop	1,000

Swap Data

Term	Rate	Use
5Y	1.30050%	TRUE
7Y	1.50200%	TRUE
10Y	1.73450%	TRUE
11Y	1.79900%	TRUE
12Y	1.84950%	TRUE
15Y	1.97200%	TRUE
20Y	2.09750%	TRUE
25Y	2.15800%	TRUE
30Y	2.19800%	TRUE
40Y	2.21520%	TRUE

mirSetUpBasisCurve (CurveCollection, StaticDataTable, CurveBuildProperties, BasisSwapConventions, BasisSwapMarketData, IndexName, [ForwardFXConventions], [ForwardFXMarketData], [SpotFXMarketData])

mirSetUpBasisSwapCurve

mirSetUpBasisSwapCurve 3M6MBasis BasisSwap Curve has been set (0.01500 sec)
 Prior to building the Basis Curve must construct an OIS Curve using 'mirSetUpOISCurve' and a Swap Curve using 'mirSetUpSwapCurve'

Curve Names

CurveCollection	EURYC
StaticDataTable	3M6MBasis
IndexName(s)	3M6MBasis:EUR3ML

<- This is a reference to a property file table and should remain fixed.

Curve Build Properties

Curve Build Properties

AsOfDate	19-Feb-16
Currency	EUR
basis.interpolation.3m6mbasis	fn_MonotoneConvexInterpolation
basis.interpolation.1m3mbasis	fn_MonotoneConvexInterpolation
basis.interpolation.6m12mbasis	fn_MonotoneConvexInterpolation

Interpolation Options

fn_LinearInterpolation
fn_StepInterpolation
fn_MonotoneConvexInterpolation
fn_SplineInterpolation
fn_ConstrainedSplineInterpolation

Example Tenor-Basis Curve II

Tenor Basis Conventions (3 X 6)		Tenor Basis Spreads (3 X 6)	
Known Leg	MarketName	EUR3M6MBASIS	
	IsLeg1Spread	FALSE	
	Leg1cashlet.calendar	TGT	
	Leg1cashlet.daycount	Act/360	
	Leg1cashlet.frequency	Semi-Annual	
	Leg1cashlet.slidingrule	MOD_FOLLOWING	
	Leg1cashlet.spotlag	2	
	Leg1discount	OIS	
	Leg1forecast	EUR6ML	
	Leg1index.accessary	6M	
	Leg1index.daycount	Act/360	
	Leg1index.fixing.calendar	TGT	
	Leg1index.frequency	Simple	
	Leg1index.payment.calendar	TGT	
	Leg1index.resetlag	2	
	Leg1index.slidingrule	MOD_FOLLOWING	
Leg2Forecast Unknown : STD is the initial guess	Leg2cashlet.calendar	TGT	
	Leg2cashlet.daycount	Act/360	
	Leg2cashlet.frequency	Quarterly	
	Leg2cashlet.slidingrule	MOD_FOLLOWING	
	Leg2cashlet.spotlag	2	
	Leg2discount	OIS	
	Leg2forecast	STD	
	Leg2index.accessary	3M	
	Leg2index.daycount	Act/360	
	Leg2index.fixing.calendar	TGT	
	Leg2index.frequency	Simple	
	Leg2index.payment.calendar	TGT	
	Leg2index.resetlag	2	
	Leg2index.slidingrule	MOD_FOLLOWING	
Solving for : Leg2Forecast Rate	Target	Leg2Forecast	
	issimultaneouseq	TRUE	
	fwdinterpolation	fn_MonotoneConvexInterpolation	
	isfwdinter	TRUE	
	issamegridindex	TRUE	
	spotrateterm	3M	
	isyieldspreadcalc	FALSE	
	Epsilon	1.00E-09	
	GradientEpsilon	1.00E-10	
	Delta	1.00E-10	
	MaxLoop	1,000	
		Term	Rate
		1Y	0.11300%
		2Y	0.11925%
		3Y	0.12425%
		4Y	0.12975%
		5Y	0.13325%
		6Y	0.13075%
		7Y	0.12725%
		8Y	0.12275%
		9Y	0.11675%
		10Y	0.11025%
		11Y	0.10425%
		12Y	0.09925%
		15Y	0.08625%
		20Y	0.06875%
		25Y	0.05825%
		30Y	0.05125%

mirSetUpBasisCurve (CurveCollection, StaticDataTable, CurveBuildProperties, BasisSwapConventions, BasisSwapMarketData, IndexName, [ForwardFXConventions], [ForwardFXMarketData], [SpotFXMarketData])

mirSetUpBasisSwapCurve

mirSetUpBasisSwapCurve

XccyBasis BasisSwap Curve has been set (0.01600 sec)

Prior to building the Basis Curve must construct an OIS Curve using 'mirSetUpOISCurve' and a Swap Curve using 'mirSetUpSwapCurve'

Curve Names

CurveCollection	EURYC
StaticDataTable	XccyBasis
IndexName(s)	XccyBasis:EURDF_USDCSA

<- This is a reference to a property file table and should remain fixed.

Curve Build Properties

Curve Build Properties

AsOfDate	19-Feb-16
Currency	EUR
basis.interpolation.xccybasis	fn_MonotoneConvexInterpolation

Interpolation Options

fn_LinearInterpolation
fn_StepInterpolation
fn_MonotoneConvexInterpolation
fn_SplineInterpolation
fn_ConstrainedSplineInterpolation

Example Xccy-Basis Curve II

Xccy Basis Swap Conventions			Xccy Basis Spreads	
Leg1Discount Unknown : OIS is the initial guess	MarketName	XCCYBASIS	Term	Rate
	adjustvalue.interpolation	fn_MonotoneConvexInterpolation	1Y	0.11300%
	isleg1spread	TRUE	2Y	0.11925%
	leg1cashlet.calendar	LNB:NYB:TGT	3Y	0.12425%
	leg1cashlet.daycount	Act/360	4Y	0.12975%
	leg1cashlet.frequency	Quarterly	5Y	0.13325%
	leg1cashlet.slidingrule	MOD_FOLLOWING	6Y	0.13075%
	leg1cashlet.spotlag	2d	7Y	0.12725%
	leg1discount	EURYC:OIS	8Y	0.12275%
	leg1forecast	EURYC:EUR3ML	9Y	0.11675%
	leg1index.accessary	3M	10Y	0.11025%
	leg1index.daycount	Act/360	11Y	0.10425%
	leg1index.fixing.calendar	TGT	12Y	0.09925%
	leg1index.frequency	Simple	15Y	0.08625%
	leg1index.payment.calendar	LNB	20Y	0.06875%
	leg1index.resetlag	2	25Y	0.05825%
	leg1index.slidingrule	MOD_FOLLOWING	30Y	0.05125%
Known Leg	leg2cashlet.calendar	LNB:NYB:TGT		
	leg2cashlet.daycount	Act/360		
	leg2cashlet.frequency	Quarterly		
	leg2cashlet.slidingrule	MOD_FOLLOWING		
	leg2cashlet.spotlag	2		
	leg2discount	USDYC:OIS		
	leg2forecast	USDYC:USD3ML		
	leg2index.accessary	3M		
	leg2index.daycount	Act/360		
	leg2index.fixing.calendar	LNB		
	leg2index.frequency	Simple		
	leg2index.payment.calendar	LNB		
	leg2index.resetlag	2		
Solving for : Leg1Discount Rate	leg2index.slidingrule	MOD_FOLLOWING		
	target	Leg1Discount		
	issimultaneouseq	TRUE		
	isfwdinter	TRUE		
	fwdinterpolation	fn_MonotoneConvexInterpolation		
	isyieldspreadcalc	FALSE		
	spotrateterm	3M		
	Epsilon	1.00E-09		
	GradientEpsilon	1.00E-10		
	Delta	1.00E-10		
	MaxLoop	1,000		

`mirSetUpFwdFXConstantCurve` (CurveCollection, StaticDataTable, CurveBuildProperties, FXForwardConventions, IndexName)

mirSetUpFwdFXConstantCurve

<code>mirSetUpFwdFXConstantCurve</code>	FwdFXConstGBP FwdFXConstant Curve has been set (0.00000 sec)
Curve Loaded	TRUE
Curve Names	
CurveCollection	EURYC
StaticDataTable	FwdFXConstGBP
IndexName(s)	FwdFXConstGBP:EURDF_GBPCSA

Curve Build Properties

Curve Build Properties	
AsOfDate	19-Feb-16
Currency	EUR

FX Forwards

FX Forward Conventions	
leg1discount	GBPYP:GBPDF
leg1forecast	GBPYP:GBPDF_USDCSA
leg2discount	EURYC:XccyBasis
leg2forecast	EURYC:EURDF_USDCSA
target	Leg2Discount

$$P(t, T)^{EUR_GBPCSA} = \left(\frac{P(t, T)^{EUR_USDCSA}}{P(t, T)^{GBP_USDCSA}} \right) \times P(t, T)^{GBP_OIS}$$

Build Order Native CSA Curves

- ① OIS Curve
- ② Standard Swap Curve(s)
- ③ Tenor Basis Curve(s)

OIS Discounting & CSA Curves

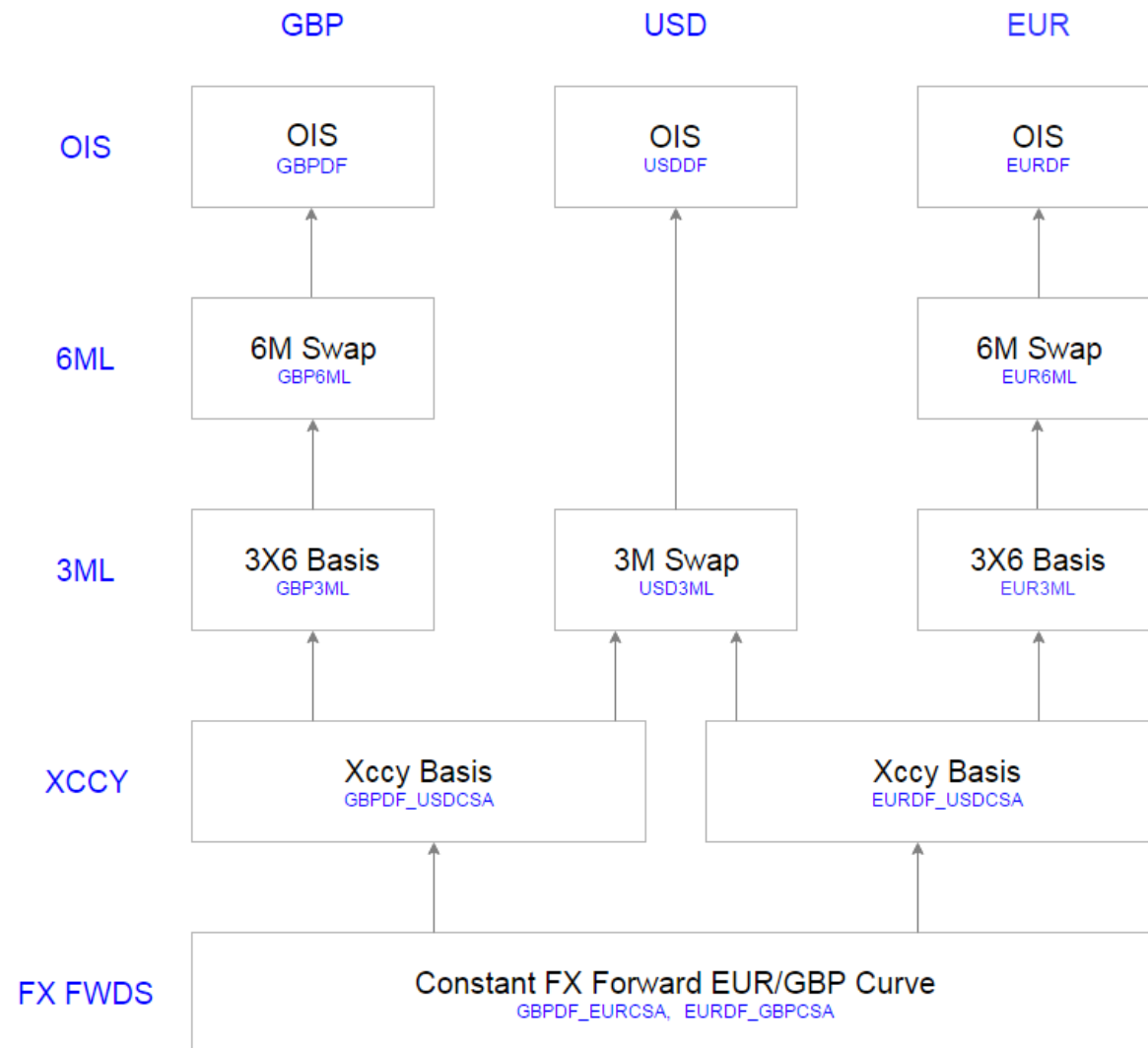
USD

	Instruments Used	USDYC
OIS	FED Funds vs 12m Fixed	OIS OIS Curve has been set (0.06200 sec)
STD	3m Float vs 6m Fixed	STD Swap Curve has been set (0.00000 sec)
3M6MBasis	Tenor Basis 3x6	3M6MBasis BasisSwap Curve has been set (0.01500 sec)
1M3MBasis	Tenor Basis 1x3	1M3MBasis BasisSwap Curve has been set (0.01600 sec)
3M12MBasis	Tenor Basis 3x12	3M12MBasis BasisSwap Curve has been set (0.00000 sec)

EUR

	Instruments Used	EURYC
OIS	EONIA vs 12m Fixed	OIS OIS Curve has been set (0.03100 sec)
STD	6m Float vs 12m Fixed	STD Swap Curve has been set (0.01500 sec)
3M6MBasis	Tenor Basis 3x6	3M6MBasis BasisSwap Curve has been set (0.00000 sec)
1M3MBasis	Tenor Basis 1x3	1M3MBasis BasisSwap Curve has been set (0.01600 sec)
3M12MBasis	Tenor Basis 6x12	3M12MBasis BasisSwap Curve has been set (0.00000 sec)

Non-Native CSA Curves have Additional Currency Dependencies



Swap Pricing, Native CSA

EUR Swaps

PV

-312,926

EUR Calibration Instruments

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	Float First	CSA	Par Rate	Present Value	PV01
1	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	1Y	1.00000%	6ML	0.00		NOCSA	-0.13969%	-11,440	100
2	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	2Y	1.00000%	6ML	0.00		NOCSA	-0.13300%	-22,791	201
3	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	3Y	1.00000%	6ML	0.00		NOCSA	-0.08700%	-32,910	303
4	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	4Y	1.00000%	6ML	0.00		NOCSA	-0.00800%	-40,682	404
5	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	5Y	1.00000%	6ML	0.00		NOCSA	0.09300%	-45,732	504
6	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	6Y	1.00000%	6ML	0.00		NOCSA	0.20200%	-48,247	605
7	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	7Y	1.00000%	6ML	0.00		NOCSA	0.32100%	-47,815	704
8	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	8Y	1.00000%	6ML	0.00		NOCSA	0.44400%	-44,630	803
9	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	9Y	1.00000%	6ML	0.00		NOCSA	0.56100%	-39,518	900
10	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	10Y	1.00000%	6ML	0.00		NOCSA	0.66700%	-33,156	996
11	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	11Y	1.00000%	6ML	0.00		NOCSA	0.76300%	-25,831	1,090
12	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	12Y	1.00000%	6ML	0.00		NOCSA	0.84700%	-18,094	1,183
13	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	13Y	1.00000%	6ML	0.00		NOCSA	0.91775%	-10,476	1,274
14	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	14Y	1.00000%	6ML	0.00		NOCSA	0.97836%	-2,951	1,364
15	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	15Y	1.00000%	6ML	0.00		NOCSA	1.03000%	4,355	1,452
16	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	16Y	1.00000%	6ML	0.00		NOCSA	1.06960%	10,704	1,538
17	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	17Y	1.00000%	6ML	0.00		NOCSA	1.10207%	16,568	1,623
18	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	18Y	1.00000%	6ML	0.00		NOCSA	1.12836%	21,911	1,707
19	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	19Y	1.00000%	6ML	0.00		NOCSA	1.14941%	26,740	1,790
20	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	20Y	1.00000%	6ML	0.00		NOCSA	1.16600%	31,069	1,872

Term	Rate	+/- Bps
2Y	-0.13300%	0.000
3Y	-0.08700%	0.000
4Y	-0.00800%	0.000
5Y	0.09300%	0.000
6Y	0.20200%	0.000
7Y	0.32100%	0.000
8Y	0.44400%	0.000
9Y	0.56100%	0.000
10Y	0.66700%	0.000
11Y	0.76300%	0.000
12Y	0.84700%	0.000
15Y	1.03000%	0.000
20Y	1.16600%	0.000
25Y	1.20100%	-
30Y	1.20400%	-
40Y	1.18600%	-
50Y	1.12000%	-

Swap Pricing, Non-Native CSA

EUR Swaps

PV

-311,136

EUR Calibration Instruments

	Effective Date	Effective Lag	Notional	Currency	Pay / Rec Fixed Rate	Maturity Or Tenor	Fixed Rate	Float Index	Float Spread	Float First	CSA	Par Rate	Present Value	PV01
1	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	1Y	1.00000%	6ML	0.00		GBPCSA	-0.13970%	-11,442	100
2	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	2Y	1.00000%	6ML	0.00		GBPCSA	-0.13305%	-22,770	201
3	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	3Y	1.00000%	6ML	0.00		GBPCSA	-0.08711%	-32,855	302
4	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	4Y	1.00000%	6ML	0.00		GBPCSA	-0.00825%	-40,589	403
5	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	5Y	1.00000%	6ML	0.00		GBPCSA	0.09252%	-45,606	503
6	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	6Y	1.00000%	6ML	0.00		GBPCSA	0.20131%	-48,105	602
7	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	7Y	1.00000%	6ML	0.00		GBPCSA	0.32012%	-47,675	701
8	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	8Y	1.00000%	6ML	0.00		GBPCSA	0.44295%	-44,513	799
9	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	9Y	1.00000%	6ML	0.00		GBPCSA	0.55990%	-39,433	896
10	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	10Y	1.00000%	6ML	0.00		GBPCSA	0.66600%	-33,103	991
11	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	11Y	1.00000%	6ML	0.00		GBPCSA	0.76222%	-25,802	1,085
12	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	12Y	1.00000%	6ML	0.00		GBPCSA	0.84647%	-18,081	1,178
13	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	13Y	1.00000%	6ML	0.00		GBPCSA	0.91748%	-10,470	1,269
14	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	14Y	1.00000%	6ML	0.00		GBPCSA	0.97836%	-2,941	1,359
15	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	15Y	1.00000%	6ML	0.00		GBPCSA	1.03030%	4,386	1,447
16	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	16Y	1.00000%	6ML	0.00		GBPCSA	1.07020%	10,769	1,534
17	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	17Y	1.00000%	6ML	0.00		GBPCSA	1.10295%	16,680	1,620
18	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	18Y	1.00000%	6ML	0.00		GBPCSA	1.12949%	22,079	1,705
19	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	19Y	1.00000%	6ML	0.00		GBPCSA	1.15076%	26,971	1,789
20	Tue 23-Feb-16	2D	1,000,000	EUR	PAYER	20Y	1.00000%	6ML	0.00		GBPCSA	1.16753%	31,367	1,872

Term	Rate	+/- Bps
2Y	-0.13300%	-0.005
3Y	-0.08700%	-0.011
4Y	-0.00800%	-0.025
5Y	0.09300%	-0.048
6Y	0.20200%	-0.069
7Y	0.32100%	-0.088
8Y	0.44400%	-0.105
9Y	0.56100%	-0.110
10Y	0.66700%	-0.100
11Y	0.76300%	-0.078
12Y	0.84700%	-0.053
15Y	1.03000%	0.030
20Y	1.16600%	0.153
25Y	1.20100%	-
30Y	1.20400%	-
40Y	1.18600%	-
50Y	1.12000%	-
		-

The End